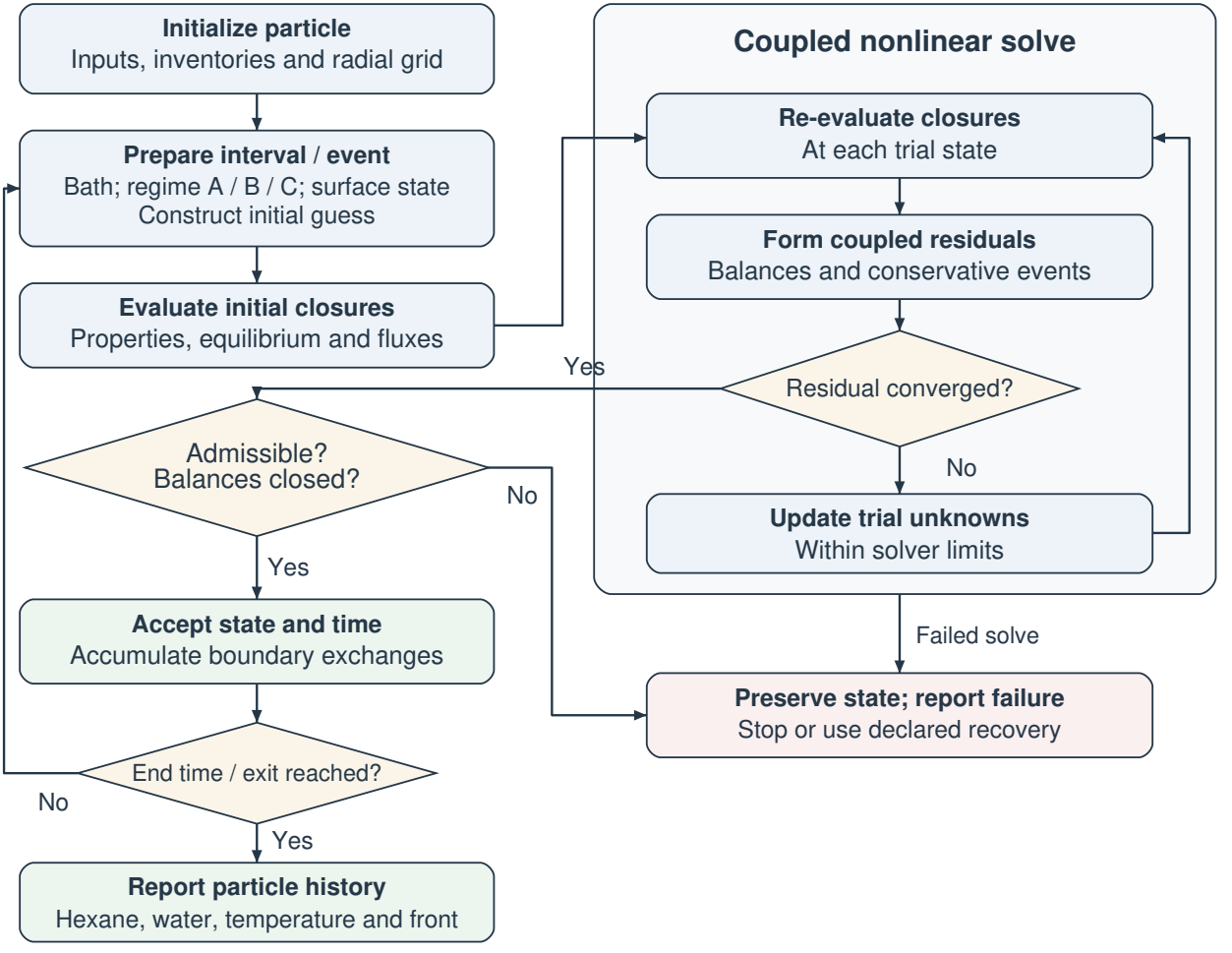

Initialize particle
Inputs, inventories and radial grid
Prepare interval / event
Bath; regime A / B / C; surface state
Construct initial guess
Evaluate initial closures
Properties, equilibrium and fluxes
Admissible?
Balances closed?
No
Yes
Accept state and time
Accumulate boundary exchanges
End time / exit reached?
No
Yes
Report particle history
Hexane, water, temperature and front
Coupled nonlinear solve
Re-evaluate closures
At each trial state
Form coupled residuals
Balances and conservative events
Residual converged?
Yes
No
Update trial unknowns
Within solver limits
Failed solve
Preserve state; report failure
Stop or use declared recovery